

Virginia Electric and Power Company
Surry Power Station
5570 Hog Island Road
Surry, Virginia 23883

SEP 19 2019

U. S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, D. C. 20555-0001

Serial No.: 19-402
SPS: SCN
Docket No.: 50-280
50-281
License No.: DPR-32
DPR-37

Dear Sir or Madam:

Pursuant to 10CFR50.73, Virginia Electric and Power Company hereby submits the following Licensee Event Report applicable to Surry Power Station Unit 1 and Unit 2.

Report No. 50-280, 50-281 / 2019-001-00

This report has been reviewed by the Station Facility Safety Review Committee and will be forwarded to the Management Safety Review Committee.

Very truly yours,



F. Mladen
Site Vice President
Surry Power Station

Enclosure

Commitment contained in this letter: None

cc: U.S. Nuclear Regulatory Commission, Region II
Marquis One Tower, Suite 1200
245 Peachtree Center Ave., NE
Atlanta, GA 30303-1257

NRC Senior Resident Inspector
Surry Power Station

IEZZ
NRR



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME

Surry Power Station, Unit 1

2. DOCKET NUMBER

05000 280

3. PAGE

1 OF 3

4. TITLE

Failure to Align Angle Drive to Pump Results in Inoperable Emergency Service Water Pump

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
07	21	2019	2019	001	00	09	19	2019	Surry Power Station, Unit 2	05000 281
									FACILITY NAME	DOCKET NUMBER
										05000

9. OPERATING MODE	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)			
N	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. POWER LEVEL	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT

Barry Garber, Manager Nuclear Licensing

TELEPHONE NUMBER (Include Area Code)

(757) 365-2725

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
A	BI	P	B260	Y					

14. SUPPLEMENTAL REPORT EXPECTED

☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO

15. EXPECTED SUBMISSION DATE

MONTH	DAY	YEAR

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On June 23, 2019, with Units 1 and 2 at 100% power, the packing gland on the 1B Emergency Service Water Pump (ESWP) failed during periodic pump testing and the pump was declared inoperable. The packing gland was repaired and the pump was returned to service on June 24, 2019. During the next scheduled pump test on July 21, 2019, another packing failure occurred. Subsequent investigation revealed that both packing failures were caused by the angle drive to pump shaft not being aligned when the pump was replaced in May 2019. The decision making was not appropriate in that only first line supervisors were consulted before making the decision not to perform the alignment. The decision should have been elevated to ensure a broader group of personnel were involved in the decision.

The ESWP 1B to angle drive alignment was performed in accordance with procedures and returned to service. Additionally, corrective actions will include a revision to the maintenance procedure and other changes focusing on improving proficiency and decision making.

This report is submitted pursuant to 10 CFR 50.73(a)(2)(i)(B) due to operations prohibited by Technical Specifications. Based on a risk assessment, the risk impact was determined to be of very low significance.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Surry Power Station, Unit 1	05000-280	2019	- 001	- 00

NARRATIVE

1.0 DESCRIPTION OF THE EVENT

Three diesel-driven vertical Emergency Service Water Pumps (ESWPs) [EIS-BI-P] are provided at the James River intake structure [EIS-KE-NN] to deliver service water (SW) makeup to the intake canal [EIS-KE-NN] in the event of a total loss of station power coincident with a design basis Loss of Coolant Accident (LOCA). The three (3) ESWPs serve both Surry units. Two of three ESWPs provide sufficient heat removal for the design basis LOCA unit and the non-LOCA unit. Technical Specification (TS) 3.14.A.4 states that when the reactor coolant system temperature or pressure exceeds 350 degrees F or 450 psig, three ESWPs shall be operable. TS 3.14.B allows one ESWP to remain inoperable for a period not to exceed 14 days.

On April 29, 2019, ESWP 1B was tagged out in preparation for pump replacement. The pump was replaced without performing an angle drive to pump alignment and on May 4, 2019, the pump was started for its return to service run. During this run, multiple vibration points were elevated. The source of these elevated vibrations were determined to be resonance and did not impact operability. On May 7, 2019, the pump was successfully returned to service. On May 8, 2019, grouting was installed under the pump pit plate and on May 28, 2019, the pump was tested with results showing that the vibrations had decreased indicating the grout changed the resonance frequency of the angle drive to pump shaft assembly.

On June 23, 2019, during periodic testing, ESWP 1B was secured due to abnormal sounds after pump start. The pump was declared inoperable as a result of a failure of the pump packing gland. The failure was attributed to loosening of the packing gland follower clamping bolts during pump operation at elevated vibration levels. Following repacking, application of a liquid threadlocker to the pump gland follower clamping nuts, and successful testing, the pump was declared operable on June 24, 2019.

During the next scheduled pump test on July, 21, 2019, ESWP 1B was again secured and declared inoperable due to a packing failure. Subsequent investigation showed that the pump shaft and angle drive were misaligned resulting in excessive shaft movement, stuffing box bushing wear, and packing failure. On July 25, 2019, the ESWP 1B angle drive to pump shaft were aligned and the pump declared operable.

It was determined that the prior packing failures were caused by angle drive to pump misalignment which occurred in May 2019. At three (3) separate times during this period of ESWP 1B inoperability (i.e., from May 7 to July 25), ESWPs 1A and 1C were separately removed from service for scheduled maintenance, but at no time was either pump unable to be returned to operable status within the 2-hour time-critical operator action starting time should a design basis event have occurred. As a result there was not a loss of safety function.

ESWP 1B was determined to be unable to perform its design function since the time that the pump was replaced and as a result exceeded its 14-day TS allowed outage time. In addition, the period of time ESWP 1A and 1C were separately removed from service resulted in exceeding the TS 3.0.1 criterion. Therefore, this report is submitted pursuant to 10 CFR 50.73(a)(2)(i)(B) as an operation or condition prohibited by Technical Specifications.

2.0 SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

The risk associated with the ESWP 1B out-of-service condition is very low since the likelihood of a design basis accident coincident with loss of off-site power and a challenge to intake canal cooling water volume is extremely unlikely. Additionally, ESWP 1A and 1C were available during the period ESWP 1B was inoperable. Therefore, the impact of the condition was very low, and there were no safety consequences because of this event.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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Surry Power Station, Unit 1	05000-280	2019	- 001	- 00

NARRATIVE**3.0 CAUSE OF THE EVENT**

This event was caused by the angle drive to pump shaft not being aligned when the pump was replaced in May 2019. The decision making was not appropriate in that only first line supervisors were consulted before making the decision not to perform the alignment. The decision should have been elevated to ensure a broader group of personnel (technical experts in vertical pump alignment and their leaders) were involved in the decision.

4.0 IMMEDIATE CORRECTIVE ACTIONS

ESWP 1B pump to angle drive alignment was performed in accordance with station procedures. The pump was tested and returned to service on July 25, 2019.

5.0 ADDITIONAL CORRECTIVE ACTIONS

Although ESWPs 1A and 1C have been in service for multiple years since their last replacement, they were inspected for extent of condition. Results showed the packing gland follower bolts/nuts were tight with no visual signs of damage. Additionally, satisfactory vibration levels and a history of successful performance runs, without gland follower damage or packing extrusion, indicated that no significant misalignment existed between the pump and angle drives.

6.0 ACTIONS TO PREVENT RECURRENCE

The maintenance procedure will be revised to provide additional guidance and require an angle drive to pump alignment when the angle drive is removed.

Other actions focusing on improving proficiency and decision making for the groups involved will be implemented by the Corrective Action Program.

7.0 SIMILAR EVENTS

None

8.0 MANUFACTURER/MODEL NUMBER

Bingham / VLTM
Johnson right angle gear drive / HH425

9.0 ADDITIONAL INFORMATION

None.